= Milestone 2 =

MusicEnjoyers

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Overview

In this project, we aim to understand how the distribution of pop music genres evolves over time, as well as which musical features become more prominent. By analyzing the changing proportions of genre populations and examining clusters of audio features across songs, we seek to uncover patterns in popular musical taste and how these preferences shift over the years.

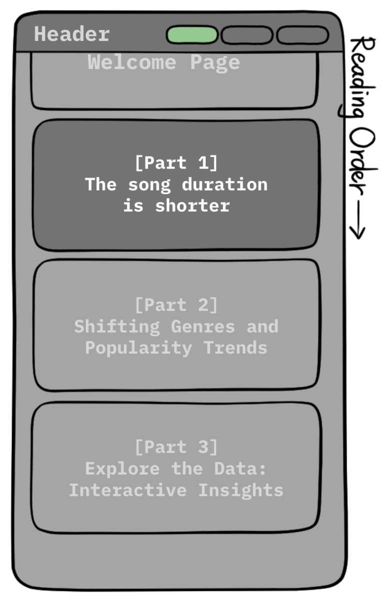
Our website consists of a welcome page followed by three main sections:

Welcome page: The big slogan “POP MUSIC IS EVOLVING”

Part1: We explore how song duration has changed over time through a line plot, highlighting the trend toward shorter tracks.

Part2: We investigate genre distribution and popularity trends, revealing how musical styles converge or diverge over time.

Part3: We present an interactive visualization: the ultimate graph that allows users to engage directly with the data and explore these patterns in a more dynamic and intuitive way.



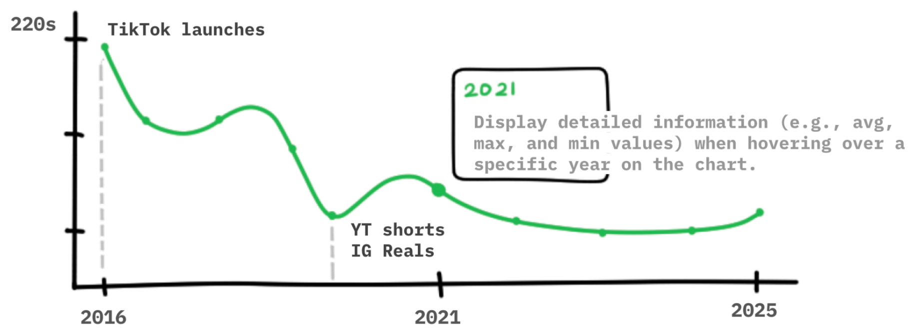
Part1: The song duration is shorter

Music listeners pursue immediate satisfaction. Platforms like TikTok, Instagram Reels, and YouTube Shorts have changed how people consume music: a song has seconds to hook the listener before they skip. Thus, music producers and artists have adapted their songs: intros are shorter and choruses come faster.

Our dataset confirms this: average song duration on Billboard Hot 100 has decreased 13 seconds from 2016 to 2025.

This answers to our core research question: **are songs getting simpler to capture attention faster?**

A prominent number highlighting the change in duration (in seconds) over the past decade.



Annotate the timeline with major events at their corresponding points in time.

The line chart shows the average song duration per year (2016–2025). X axis = year, Y axis = average duration in seconds. A tooltip on hover shows the exact duration, number of songs, the shortest song, and the longest song for that year.

A line chart is chosen because duration is a quantitative attribute over a continuous time axis, position is the highest-accuracy channel for quantitative data per Lecture 6_2 Marks & Channels.

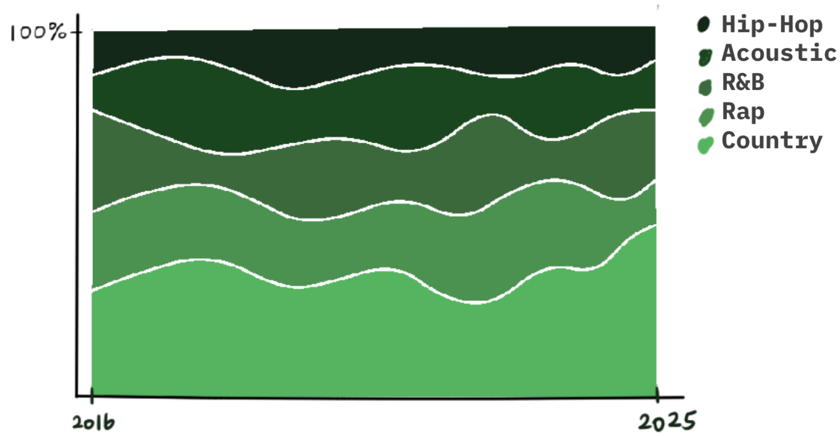
This section is implemented using [D3.js](), HTML, CSS, and Google Fonts.

- D3 for drawing the line chart (Lecture 4_2 & 5_2 D3.js + Interactive D3)
- HTML and CSS to structure the page and style all non-chart elements (Lecture 1_2 Web Development)
- Google Fonts for typography (Lecture 12_1 Storytelling)

Part2: Shifting Genres and Popularity Trends

In this section, we explore how the distribution of pop music genres changes over time and which musical features become more prominent. By analyzing the ratio of genre populations and examining clusters of audio features across songs, we aim to uncover patterns in popular musical taste and how these preferences evolve over the years.

To support this exploration, we employ a variety of visualizations. A **ratio-changing chart** illustrates how genre proportions shift over time, while an animated scatter plot reveals how songs cluster based on their audio features across different years.



This section is implemented using **D3.js** and **Canvas**, enabling both flexibility and performance for dynamic visualizations. It also draws on key course concepts, including **storytelling** and **perceptual color design**, to effectively communicate trends and patterns.

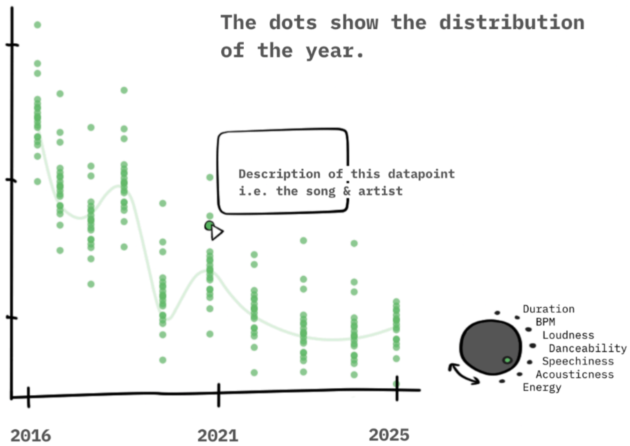
Part3: Explore the Data: Interactive Insights

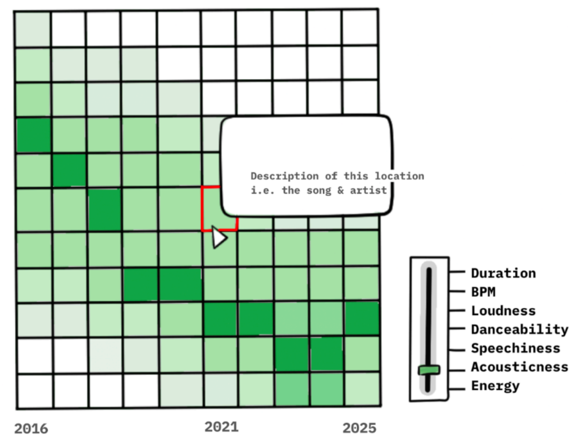
The main landing page will contain **two main views**. Each, in addition to the graph will have buttons where the graph will toggle between the following categories: Duration, BPM, Loudness, Danceability, Speechiness, Acousticness, and Energy.

In addition, each of these buttons, when hovered over for a certain duration will display a description of the category

View 1: The main view is a line plot, where the x-axis denotes the year and the y-axis denotes the average of the chosen category. When hovered over, the specific average will be displayed, similar to the first part. With this part, we can see how songs and their various features have changed over the years.

View 2: The second view is a heat map. The axes are the same as the view one. What changes as well is that when hovering over the heatmap, it will display the locations datapoints, i.e. the song and artist of said datapoint, as well as the actual value.



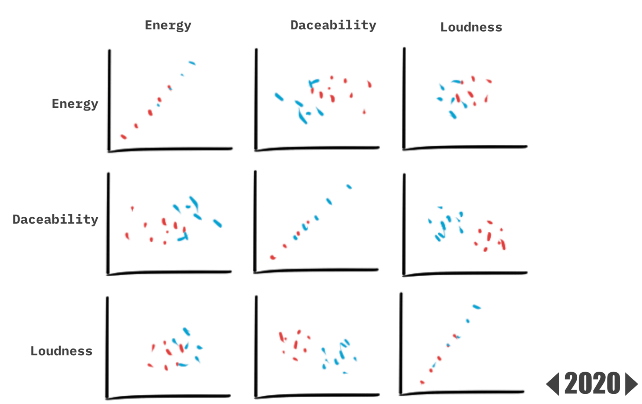


We will be using mainly D3 and JavaScript for the graphs, HTML and CSS for the Website Skeleton. We will also use Crossfilter for better interaction with users.

(Reference Lectures: JavaScript Part 1 and 2, D3, More interactive d3.js, Lecture 5_1 Interaction, Perception colors)

Extra Ideas

- In part 2, we can also incorporate additional views, such as **cluster compactness and separateness for storytelling**, **feature heatmaps**, and **multiple coordinated views**, to provide a richer and more interactive understanding of the data.
- In part 3, we can make an additional toggle for various music genres for the heatmap.
- Also, we would like to explore lyrics complexity, emotional tone, and thematic topics over time, focusing on lyrical content rather than musical features or melodies, if time permits.



Use multiple coordinated views to present multivariate clustering results of a year.